Smart Parking Platform

Project Management Plan — Weekly Deliverables

Final Project Due: December 4, 2026

Week 1: Research, Vision and SRS

Research and project idea

Our original research focused on ParkMobile, Flowbird and ParkWhiz as examples of parking apps. For our project, we want to bring the driver's experience and parking management tools together in one simple platform.

WE ARE $oftware ¢orp. is the fictional company in our project scenario. The goal is to help drivers find and reserve parking while giving operators an easier way to manage their facilities.

Vision and scope

Drivers will be able to search for parking, check availability and prices, reserve a space, make a test payment, and review their bookings. Operators will update facility details and view occupancy and booking summaries.

For the semester, we will focus on a mobile-friendly web prototype with sample parking data, simulated availability and test payments. Live sensors, real payments, and separate native mobile apps are outside this release.

SRS

Area	System requirement
Accounts	Register, sign in/out, reset a password, and edit a profile. Keep driver and operator's permissions separate.
Parking search	Search by destination and show location, hours, price, and availability. Keep the address visible if the map fails.
Availability	Show simulated updates within 30 seconds. Warn users and block new bookings if the fee is over 60 seconds old.
Reservations	Create a five-minute payment hold; prevent double booking and release expired holds.
Payments and history	Use sandbox payments, confirm each booking only once, and show the driver's booking history and test receipts.
Cancellation and updates	Allow cancellation before the start time, release the space, and record a test refund. Provide in-app status messages and a way to report incorrect data.
Operator tools	Let assigned operators update facility details, rates and occupancy, and view daily booking summaries.
Quality requirements	Target search/detail responses within 2 seconds for 95% of requests in a 25-user test. Check keyboard access, phone/desktop layouts, permissions and backup recovery.

Week 2: Work Breakdown Structure

WBS	Deliverables and work packages
1.0	Smart Parking Platform
1.1	Research and planning 1.1.1 Review parking apps and define scope 1.1.2 Prepare the schedule, budget and management plan
1.2	Requirements 1.2.1 Define driver and operator needs 1.2.2 Update the SRS, stories and acceptance checks
1.3	System design 1.3.1 Design architecture and APIs 1.3.2 Design database and sample data 1.3.3 Design screens and user flows
1.4	Application development 1.4.1 Accounts and permissions 1.4.2 Search, maps and availability 1.4.3 Reservations, cancellation and history 1.4.4 Test payments and refunds 1.4.5 Directions and notifications 1.4.6 Operator tools and reports
1.5	Integration 1.5.1 Connect the interface, API and database 1.5.2 Connect maps and sandbox payments 1.5.3 Check inventory and booking events
1.6	Testing 1.6.1 Functional and regression tests 1.6.2 Security and performance checks 1.6.3 User acceptance testing and defect fixes
1.7	Release and handoff 1.7.1 Prepare the demo and test backup recovery 1.7.2 Complete the user guide and known limitations
1.8	Closeout 1.8.1 Submit the final project 1.8.2 Review lessons learned

Ownership and task tracking

The PM maintains the plan, the BA maintains requirements, and the technical leads own implementation tasks. Each task should have one lead, a clear output and a status on the team board.

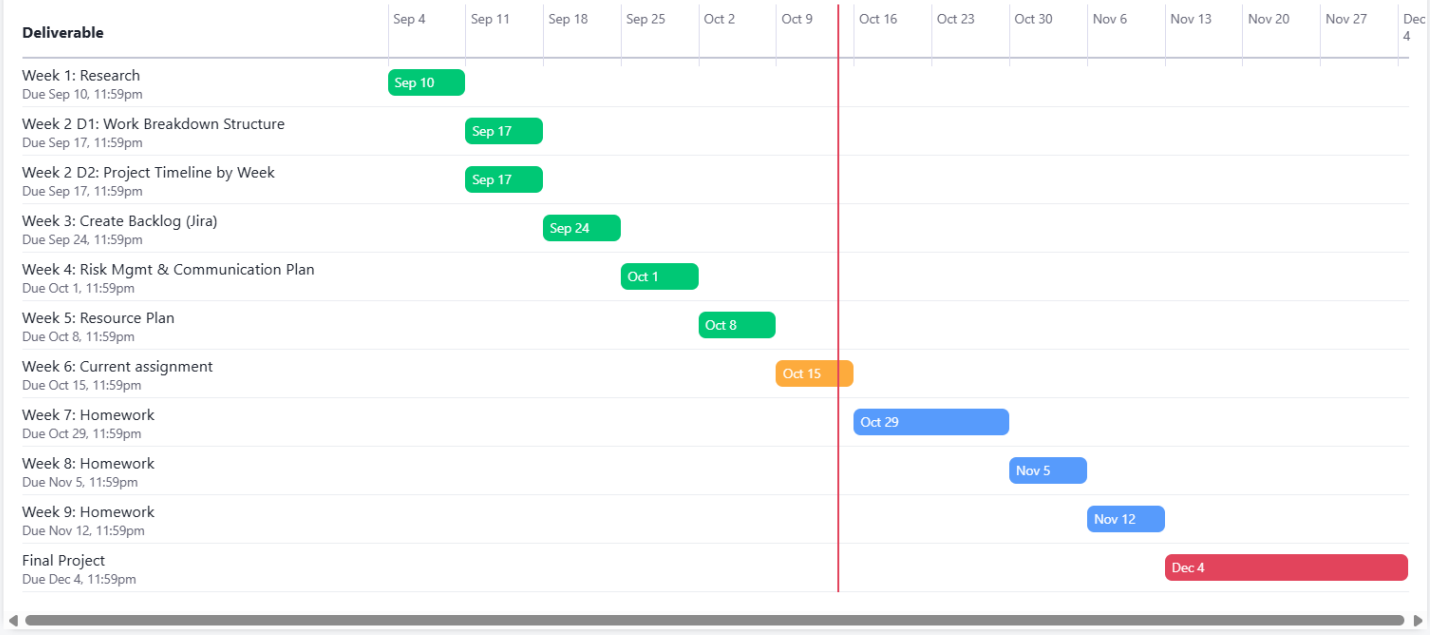
Week 2: Timeline and Gantt Chart

Homework runs weekly through Week 9. The work after November 5 is final project preparation, not additional weekly homework.

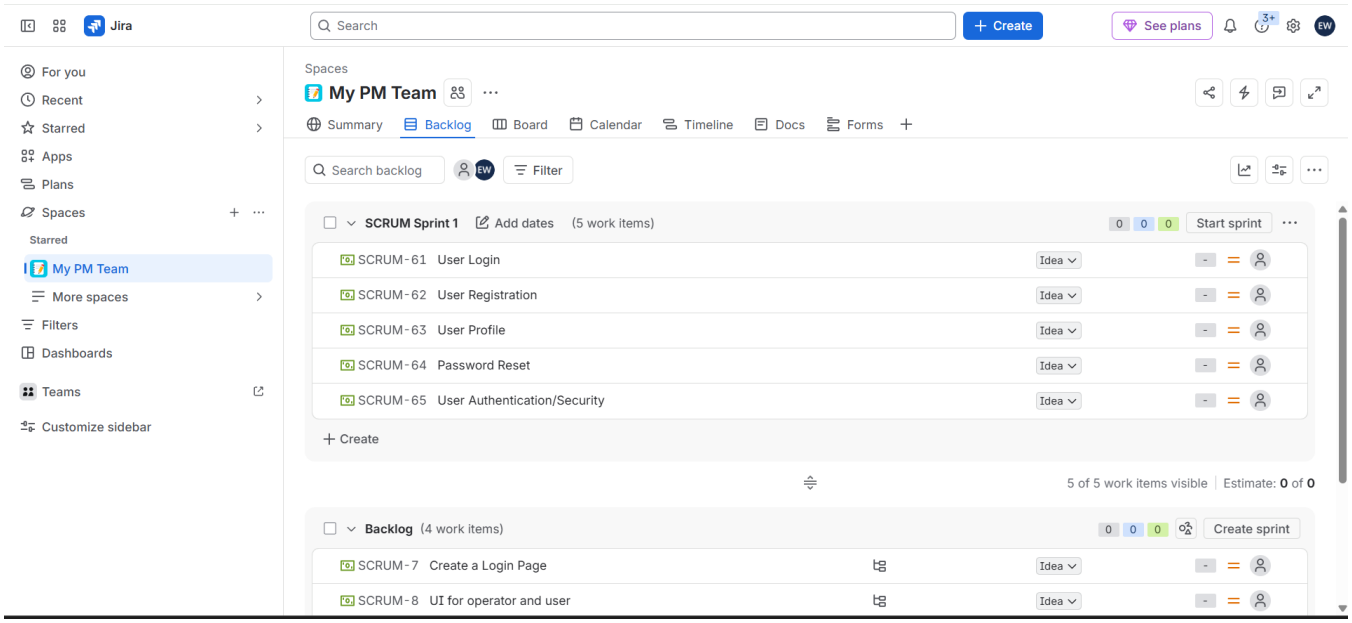
Smart Parking Platform

Project Management Plan: Weekly Deliverables · Final due Dec 4, 2026

Past due date | Current (Week 6) | Upcoming | Final project / today line



Week 3: Create Backlog (Jira)



Week 4: Risks and Communication

Updated risk register

Risk	P / I	Owner	Prevention and response
Wrong or stale availability	M / H	BE	Timestamp updates and test concurrent bookings. Block reservations when data is stale.
Map or location failure	M / M	FE	Check facility coordinates and provides a readable address fallback.
Payment failure or duplication	M / H	BE	Test failures and repeated callbacks. Do not confirm a failed payment; reconcile test refunds.
Security or privacy defect	M / H	SEC	Test permissions and keep secrets out of code. Disable the affected feature until fixed.
API/cloud costs rise	M / M	PM	Track usage and set limits. Switch to a simulator or approve reserve use if needed.
Absence or uneven workload	M / M	PM	Share notes and review capacity. Reassign blocked work early.
Late integration or defects	H / H	QA / BE	Integrate and test each sprint. Pause new work to fix release-blocking defects.
Scope changes or deadline pressure	M / H	PM / BA	Review impact before approval. Defer lower-priority work and protect the final week.

Communication plan

Audience/purpose	Frequency and method	Owner
Whole team: progress, tasks and blockers	Weekly meeting; short team-chat check-ins three times a week	PM
Technical team: interfaces, bugs and tests	Weekly or when blocked; board and focused chat/call	BE / QA
Backlog and sprint planning	Weekly refinement; planning/review every two weeks	BA / PM
Budget and risks	Weekly check; shared cost log and risk register	PM
Instructor: progress and feedback	Every two weeks or as required; short report or presentation	PM
Urgent issue or scope decision	Same working day; team chat plus a recorded board decision	PM

Week 5: Roles, Cost Plan and RACI

Roles and resource plan

Role	Name	Hours	Rate	Cost
Project Manager	Max Miller	60	$40	$2,400
Business Analyst	Sarah Thomas	60	$35	$2,100
UI/UX Designer	Michael Davis	100	$35	$3,500
Front-End Developer	David Brown	200	$40	$8,000
Back-End Developer	Emily Gabriel	200	$45	$9,000
Database Administrator	Alex Martinez	80	$40	$3,200
Tester	Olivia Garcia	80	$30	$2,400
Security Specialist	Jose Martinez	40	$45	$1,800
Total		820		$32,400

Cost plan

Item	Estimate
Labor: 820 hours	$32,400
Cloud hosting: 3 months × $200	$600
Maps/API allowance: 3 months × $150	$450
Software and testing tools	$300
Base estimate	$33,750
Contingency: 10%	$3,375
Total project estimate	$37,125

RACI matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed. UX = design; DB = database; SEC = security. Each row has one accountable role.

Task	PM	BA	UX	FE	BE	DB	QA	SEC
Planning and budget	A/R	R	I	C	C	I	C	I
Requirements	C	A/R	C	C	C	C	C	C
Design	I	C	A/R	R	C	C	C	I
Implementation	I	C	C	R	A/R	R	C	C
Testing and security	I	C	I	R	R	C	A/R	R
Release and submission	A	R	I	R	R	R	R	C

Week 6: Project Update

What was updated

Area	Week 6 update
SRS and user stories	Clarified the core features, added quality requirements, and linked the backlog to user outcomes.
WBS and schedule	Kept the main work packages and added a clearer sprint schedule, dependencies, and Gantt chart.
Backlog and sprints	Organized 16 core stories into four proposed sprints and separated future features from the semester release.
Risks and communication	Added owners, practical responses, meeting frequency, and escalation rules.
Roles, costs and RACI	Kept the original scenario team and budget, checked the totals, and clarified accountability.